LifeMate: from Vietnamese speech to tool actions

ZAIC 2026 / Training, offline inference, and release checks

01 FIT THE AGENT ON MOLAB

Official TRAIN split
1,158 fitting requests

→ **Build action targets**
3,801 calls and finish responses

→ **Fine-tune Qwen3-4B**
Rank-16 LoRA; step 362 saved

→ **Merge adapter for eval**
Pinned base and adapter

02 RUN EACH SAMPLE OFFLINE

Audio and history
Vietnamese request and prior turns

→ **Whisper large-v3-turbo**
Transcribe user audio

→ **Chronological prompt**
History, current request, context and installed schemas

→ **Qwen tool agent**
Propose one call or finish

Tool call →

← Tool result

Official ToolEnv
One per sample; validate and execute

Finish →

Export one JSONL row
ID, complete call trace, state hash

Historical calls are context. The agent executes only calls for the current request. Stop within the sample budget and export the trace actually produced.

03 COMPARE, FREEZE, AND PACKAGE

Audio pilot (48)
LoRA 19.35 vs 12.85 baseline

→ **Development (232)**
Paired score and timing

→ **Freeze and evaluate**
155 holdout examples

→ **Generate and verify**
634 public IDs
Docker and timing notebook

Pilot scores are local estimates. LoRA advancement depends on the paired development result and the timing limit.